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Is the Incidence of Concomitant Esotropia with Diplopia in the United States Increasing?

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Purpose: To investigate the incidence of comitant esotropia (ET) with diplopia using a large insurance claim database in the United States.

Design: A retrospective cross-sectional study.

Subjects: A large health insurance database.

Methods: Claims for comitant ET with diplopia were extracted using a combination of ET- and diplopia-related codes. The claim ratio of ET to strabismus was calculated from 2007 to 2022. Age and sex distributions were compared between the first (2007–2014) and second (2015–2022) half periods.

Main Outcome Measures: Change of ratio of ET with diplopia to all strabismus claims.

Results: In total, 38 404 ET claims were identified among 1 516 779 strabismus claims. The mean age of the ET claims was 39.0 ± 17.4 years. A more marked female predominance in ET claims (male:female = 1:1.6) than in strabismus claims (1:1.1) was observed. The ET claim ratio linearly increased from 1.27% in 2007 to 3.49% in 2022 (adjusted $R^2 = 0.856$, $P < 0.001$). The mean age of the ET claims was lower in 2015 to 2022 than in 2007 to 2014 (38.4 vs. 40.0 years, $P < 0.001$). The female predominance was more marked during 2015 to 2022 than in 2007 to 2014 (1:1.8 vs. 1:1.3, $P < 0.001$). The sex distribution of the younger age group (<39 years) changed from male predominance in 2007 to 2014 to female predominance in 2015 to 2022.

Conclusions: The proportion of comitant ET with diplopia in strabismus has increased since 2007 in the United States. The increase was most pronounced in young patients. Female predilection in comitant ET with diplopia was observed, and the change from male to female predominance over time was remarkable among young patients.

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Comitant esotropia (ET) with diplopia is an acquired form of ET complicated by diplopia. This type of ET includes acquired ET, divergence-insufficiency ET, convergence-insufficiency ET, convergence-excess ET, and sagging eye syndrome. Some studies have reported an increased incidence of this disease based on single-center observations.[1,2] Although the disease was considered rare,[3] a recent increase in the disease poses public health concerns and raises some suspicion that different etiologies of the disease may be responsible. To date, this increasing tendency has only been investigated in single centers; thus, these findings cannot be generalized.

This study investigated the incidence of concomitant ET with diplopia using a large insurance claim database in the United States. Additionally, we investigated the changes in age distribution and sexual preference for the entity. Our findings may contribute clues to the underlying mechanisms.

Methods

This study was conducted using the deidentified Merative MarketScan Commercial Database (2007–2022).[4] The MarketScan Database includes health insurance claims across the care continuum (e.g., inpatient, outpatient, outpatient pharmacy, and carve-out behavioral health care), as well as enrollment data from large employers and health plans across the United States. The data for this project were accessed from the Stanford Center for Population Health Sciences Data Core. The Population Health Sciences Data Core was supported by the National Institutes of Health National Center for Advancing Translational Science Clinical and Translational Science Award (UL1TR003142) and internal Stanford funding. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Insitutes of Health. This research was conducted under protocol 40974 and approved by the Stanford Administrative Panel on Human Subjects Research (Institutional Review Board). This study adhered to the principles of the Declaration of Helsinki. The need for informed consent was waived by the institutional review board for using only deidentified databases.

A study flowchart is shown in Figure 1. The diagnoses of the individual claims were identified using the International Classification of Diseases, Ninth and Tenth Revisions (ICD-9-Clinical Modification [CM] and ICD-10-CM, respectively). Claims that included the diagnostic codes for strabismus (378x in ICD-9-CM; H49x to 52x in ICD-10-CM) were selected. Claims made at the age of <7 years were excluded. The database included only those aged ≤65 years. Thus, the claims data were confined to individuals aged 7 to 65 years. We classified these claims as

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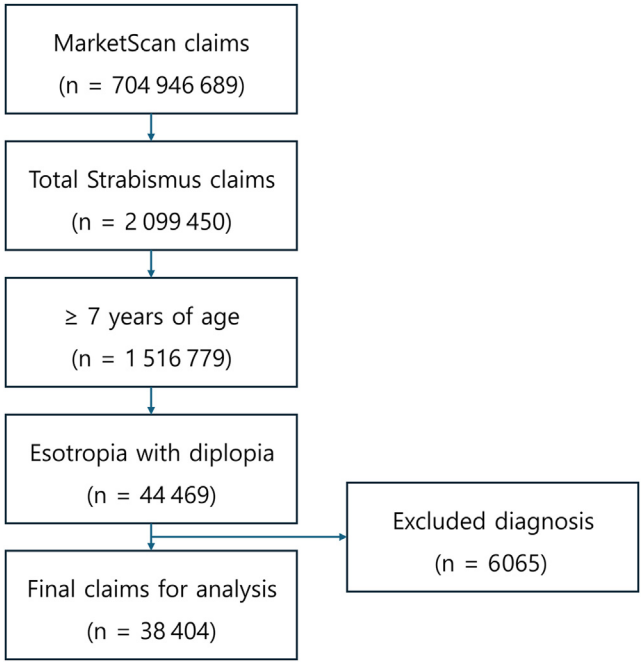


Figure 1. Study flowchart of data selection from the MarketScan Database.

concomitant ET with diplopia when the claims were coded with ET-related codes (3780x, 37821, 37822, 37835, 37841, 37884, 37885 in ICD-9-CM; H500, H5031, H5032, H5043, H5051, H5112 in ICD-10-CM) and diplopia codes (3682 in ICD-9-CM; H532 in ICD-10-CM) at the same visit. We excluded claims with diagnostic codes for paralytic, neurological, mechanical, and congenital strabismus, as listed in Table S1 (available at www.ophthalmologyscience.org).

The ratio of comitant ET with diplopia to total strabismus claims was calculated annually to investigate the annual changes in this ratio. The entire period covered by the MarketScan Database was divided into the first (2007–2014) and the second half periods (2015–2022). We compared the age and sex differences between the 2 time intervals. We used the mean age of ET claims (39 years) as the cutoff, and the ET claims were divided into 2 groups: younger (<39 years of age) and older (≥39 years of age). The sex ratio differences of the different age groups were determined.

Statistical significance was set at $P < 0.001$, considering the large sample size. The linear tendency of the yearly comitant ET with the diplopia ratio was evaluated using a linear regression model. Categorical differences between the groups were assessed using the chi-square test. Age differences between the groups were tested using the independent *t* test. Python version 3.12.7 (https://www.python.org/) was used for statistical analyses and model construction. Data preprocessing was performed using the Pandas package (version 2.2.3; https://pandas.pydata.org/). The SciPy library (version 1.14.1; https://scipy.org/) was also used for statistical analyses. Additionally, a linear regression model was constructed using the Statsmodels library (version 0.14.4, https://www.statsmodels.org/).

Results

This study included 1 516 779 strabismus-related claims that spanned between 2007 and 2022. Comitant ET with diplopia was identified in 38 404 claims (2.53% of all the strabismus claims). The sex ratio (male to female) of strabismus was 1:1.1 in total, and comitant ET with diplopia was 1:1.6. A female predominance was observed for concomitant ET with diplopia. The mean age at which the claims were made for concomitant ET with diplopia was 39.0 years. This reflects an older age than the mean age of total strabismus (mean age, 35.7 years; $P < 0.001$, independent *t* test) (Table 2).

The ratio of comitant ET with diplopia to total strabismus claims consistently increased over time, from 1.27% in 2007 to 3.49% in 2022. The linear regression model showed strong goodness of fit (adjusted $R^2 = 0.856$, $P < 0.001$), which meant 85.6% of the increasing trend could be predicted using the linear regression model. The ratio increased by approximately 13.9% of 2007's ratio every year (coefficient for yearly increment was 0.1389, $P < 0.001$) (Fig 2).

The mean age of comitant ET with diplopia claims was lower during 2015 to 2022 than in 2007 to 2014 (38.4 vs. 40.0 years of age, $P < 0.001$, independent *t* test, Table 2). The proportion of claims in the younger age group was higher during 2015 to 2022 than in 2007 to 2014 (48.8% vs. 42.3%, $P < 0.001$, chi-square test). The claim ratio during 2015 to 2022 was higher among individuals in their 20s to 30s and lower in those in their 50s compared with the ratio during 2007 to 2014 (Fig 3).

The 2015 to 2022 interval displayed a substantially more marked female predominance than the 2007 to 2014 interval (male:female ratio, 1:1.8 vs. 1:1.3, chi-square test, $P < 0.001$, Table 2). The younger age group showed a greater male predominance during 2007 to 2014. In contrast, during 2015 to 2022, a female predominance was observed. Furthermore, sex predilection remained similar from 2007 to 2022 in the older age group (Fig 4).

Table 2. Baseline Characteristics of Total Strabismus and ET with Diplopia Claims

Characteristics	Strabismus Total	ET with Diplopia			
		Total	*2007–2014 (1)*	*2014–2022 (2)*	*1 vs. 2*
n	1 516 779	38 404	13 583	24 821	
Sex (male:female)	1:1.1 (718 911:797 868)	1:1.6 (14 753:23 651)	1:1.3 (5944:7639)	1:1.8 (8809:16 021)	<0.001*
Age					
Mean	35.7	39.0	40.0	38.4	<0.001†
SD	19.5	17.4	17.8	17.2	
min	7	7	7	7	
max	65	65	64	65	

ET = esotropia; max = maximum; min = minimum; SD = standard deviation.
*Chi-square test.
†Independent *t* test.

Discussion

Using the MarketScan Database, this study revealed an increasing tendency of comitant ET with diplopia-related claims from 2007 to 2022. Comitant ET with diplopia demonstrated a significant female predominance than other strabismus claims. The mean age of patients was lower during 2015 to 2022 than during 2007 to 2014. A stronger female predilection over time was found among the younger patients. In contrast, female predominance remained stable in the older patients.

An increasing incidence of comitant ET with diplopia has been suspected, and some recent studies have demonstrated these tendencies. Kim et al reported an increased incidence of acquired ET in a single strabismus clinic in Korea. They compared the incidence of new cases of strabismus with new intermittent exotropia cases.[2] The authors reported the ratio of acquired ET to total strabismus increased from 0.14% (1/737) in 2008 to 0.87% (14/1613) by 2021. Chen et al reported that the proportion of divergence-insufficiency ET in acquired ET increased from 1978 to 2018 in a single provider's practice.[1] They reported that the divergence insufficiency to acquired ET ratio almost tripled from 1978–1998 to 1999–2018 (11.8%–29.4%).

Consistent with the findings of previous studies, our large dataset confirmed that comitant ET with diplopia had

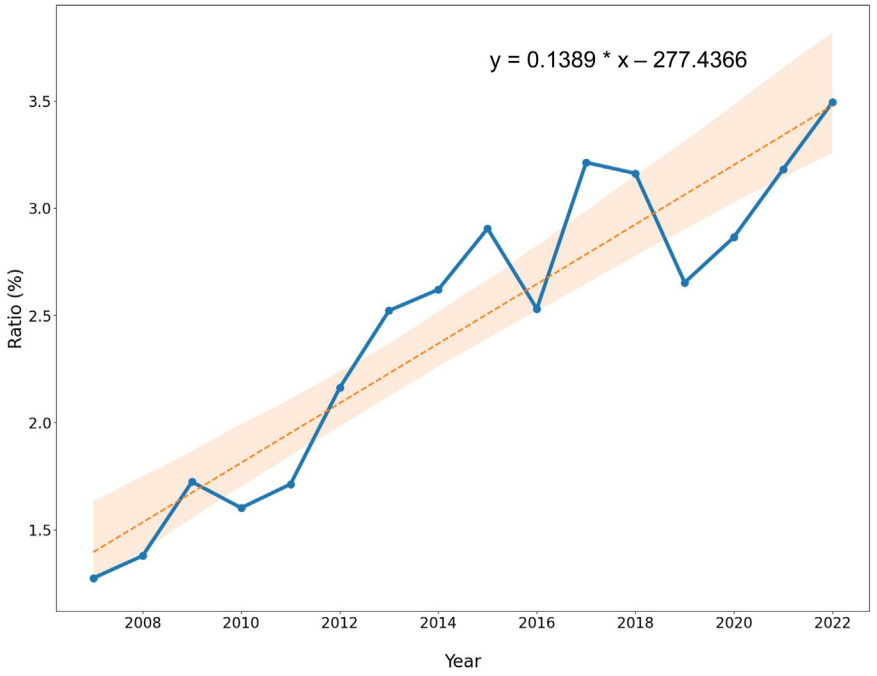


Figure 2. The annual claim ratio changes of acquired esotropia to total strabismus (depicted in solid line and dots) and its linear regression model (depicted in dashed line and shade [95% confidence interval], adjusted $R^2 = 0.856$).

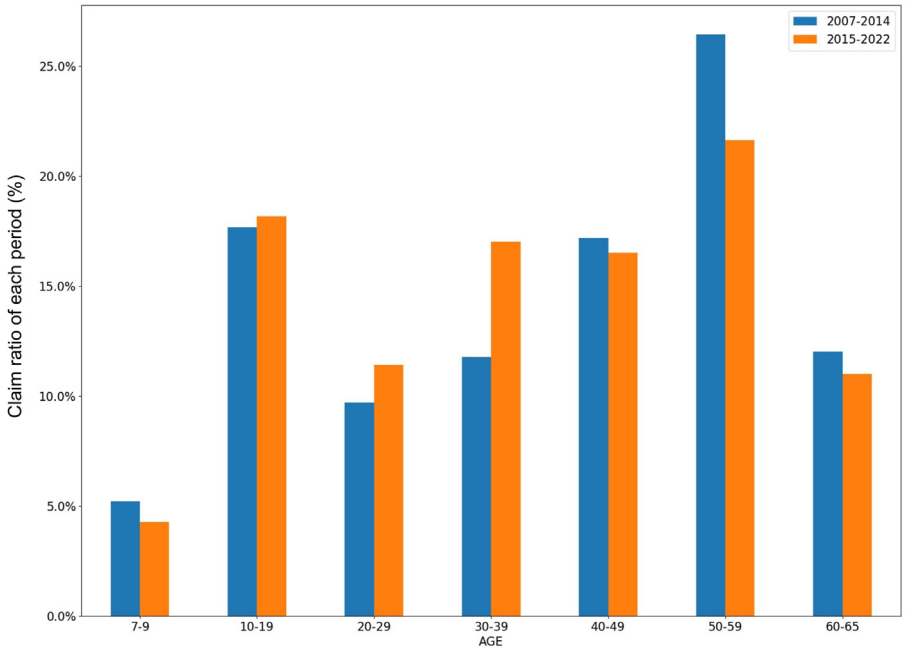


Figure 3. Age-specific claims during 2007 to 2014 and 2015 to 2022.

increased from 2007 to 2022. Many early reports focused on the possibility of a neurological etiology for the acquired[5–10] and divergence insufficiency of ET,[11,12] considering the disease as a rare entity. However, more recent studies have reported many cases without neurological problems.[2,13–15] These findings suggest that the number of patients with newly emerging causes of mechanical imbalance of the extraocular muscles or orbital tissues,[15–18] apart from previously suspected causes, has increased.

One noteworthy finding of this study was that the age distribution of patients differed between the 2 time intervals

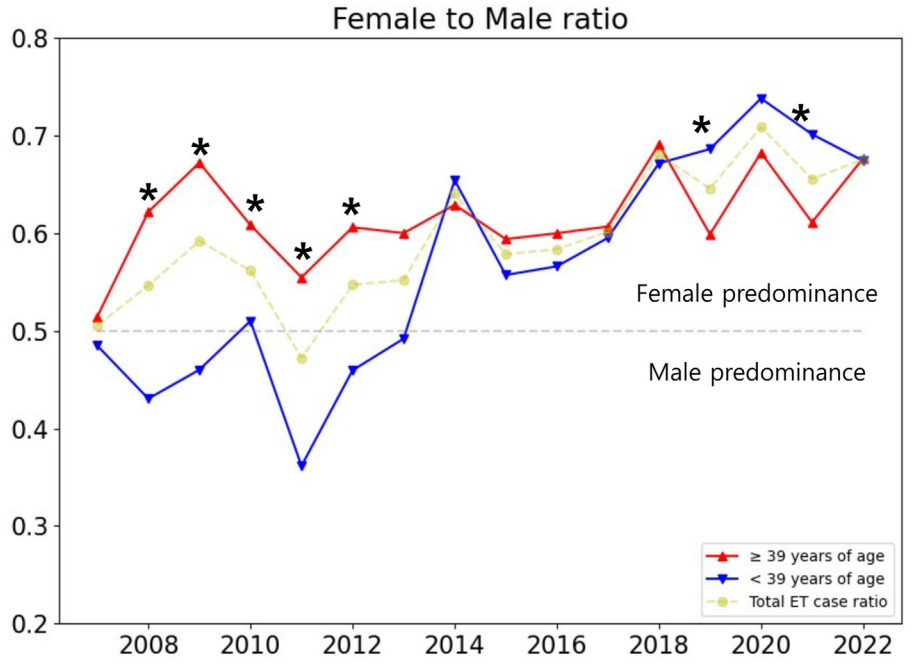


Figure 4. Annual female-to-male ratio changes between younger (<39 years) and older (≥39 years) age groups. Ratio >0.5 and <0.5 means female and male predominance, respectively (*: $P < 0.001$, chi-square test for sex distribution differences between the age groups). ET = esotropia.

(2007−2014 vs. 2015−2022). The proportion of younger patients, especially those in their 20s and 30s, was higher during 2015 to 2022 than in 2007 to 2014. This finding suggested that the increased incidence of the comitant ET with diplopia was led by the younger patients. Younger patients may be more susceptible to emerging causes. Still, another possibility for this finding is younger patients tended to seek treatment sooner than older patients in the latter half period.

In this study, we found that patients with comitant ET with diplopia showed far more female predominance than the entire strabismus population, while the sexual preference of this disease entity differed from study to study (male[2,13,14,19,20] vs. female predominance[18,21,22]) based on a single clinic or hospital setting. The female predominance intensified further during 2015 to 2022. More importantly, the sex predilection changed among the younger patients, from male to female, suggesting an intensification of female preference for the entity. In contrast, female predilection remained relatively stable among older patients throughout the study period. Taken together, our findings suggest that younger female patients with concomitant ET with diplopia may be more susceptible to new causes.

Many studies have speculated that the sudden surge in the incidence of concomitant ET with diplopia may be due to excessive near-work related to the use of smartphones,[18] laptops, or refraining from the use of progressive addition lens.[17] These speculations commonly insist that excessive near work may cause accommodative convergence, resulting in increased tonus of the medial rectus and weakening of the lateral rectus, and that these imbalances may induce ET. Given that smartphones have become an essential part of daily life, an increase in near work has become unavoidable. These etiological speculations are gaining considerable credibility. One study reported that women tended to use smartphones for a longer duration than men (166.78 minutes vs. 154.26 minutes per day). The same study also reported that teenagers tended to use smartphones for longer than any other age group and that the usage time decreased as age increased.[23] These results are consistent with the findings of this study, in particular, the sexual predilection and age distribution of concomitant ET with diplopia during 2015 to 2022. However, the exact association between smartphone use and comitant ET and diplopia remains unclear.

This study has some limitations. First, studies using diagnostic codes in claims databases are not able to extract the exact diagnoses. Our study was not able to evade these issues, even though we excluded the codes unrelated to comitant ET with diplopia as much as possible. The actual incidence of comitant ET with diplopia may be different from our results. Second, comitant ET with diplopia includes several disease entities. Hence, demographic results, such as age distribution or sexual preference, may differ depending on the demographics of each individual diagnosis. Third, we investigated the relative ratio of concomitant ET with diplopia to total strabismus claims. The number of claims from the MarketScan Database differs from year to year; therefore, the ratio increase does not necessarily equate to increased claims for the disease.

In conclusion, the proportion of comitant ET with diplopia in strabismus has increased since 2007 in the United States. The increase was the most prominent among young patients. This disease entity demonstrated a female predilection, and this tendency has intensified over time. Furthermore, the change from male to female predominance over time was remarkable in young patients.

Data Statement

The data for this project were accessed from the Stanford Center for Population Health Sciences (PHS) Data Core. The PHS Data Core was supported by the National Institute of Health National Center for Advancing Translational Science Clinical and Translational Science Award (UL1TR003142) and internal Stanford funding. The content represents the views of the authors and does not necessarily reflect the official views of the National Insitutes of Health. Certain data were supplied by Merative as part of one or more MarketScan Research Databases. Any analysis, interpretation, or conclusion based on these data is solely that of the authors and not Merative.

Footnotes and Disclosures

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Disclosures:

All authors have completed and submitted the ICMJE disclosures form.

The authors made the following disclosures:

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HUMAN SUBJECTS: Human subjects were included in this study. This research was conducted under protocol 40974 and approved by the Stanford Administrative Panel on Human Subjects Research (Institutional Review Board [IRB]). This study adhered to the principles of the Declaration of Helsinki. The IRB waived the requirements to obtain informed consents in this study.

No animal subjects were used in this study.

Author Contributions:

Conception and design: Kim, Lambert

Data collection: Kim, Lambert
Analysis and interpretation: Kim, Lambert
Obtained funding: N/A
Overall responsibility: Kim, Lambert

Abbreviations and Acronyms:
CM = clinical modification; **ET** = esotropia; **ICD** = International Classification of Diseases.

Keywords:
Esotropia, Diplopia, Insurance claim, Epidemiology.

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